

3D PRINTING Quickens Manufacturing, Customises Healthcare, And More



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There is a lot you can do with a 3D printer. You can create doll houses and toys for your children, or decorative objects for your home. You can print chocolate in various shapes. Real food, like pizzas, can be printed in outer space, auto components can be economically and quickly printed, and limbs and other implants hastily created. Although most 3D printers are custom-made or sold with a heavy price tag today, things are on the verge of changing. In the coming years, a 3D printer will become one of those objects that we will own—simply because it does exciting things! How exciting? That's the question we attempt to answer now.

Made for space, in space

Building things in outer space is a huge challenge as spaceships cannot carry the weight of equipment along with the raw materials. The way out is to 3D print all the stuff in outer space itself. California-based Made in Space (MIS) has shown that it is possible.

MIS owns and operates the Additive Manufacturing Facility (AMF)—a fully-functional 3D printing set-up that reached the International Space Station (ISS) in

March 2016, and has since been producing various parts for MIS' customers. But, why would one want to produce parts in outer space? The company believes that the space environment is unique, with its persistent microgravity and vacuum conditions. Manufacturing and assembling components in space widens design possibilities and enables one to extend the life of existing on-orbit assets or entirely new structures.

Take the example of optical fibre. The company claims that when exotic optical fibre is produced in a strong gravitational environment, such as Earth, it is subject to some flaws. Tiny crystals appear in the fibre and significantly increase signal loss.

According to them, "In space, ZBLAN optical fibre can be produced without these crystals, providing superior data transmission capabilities compared to both Earth-produced ZBLAN and traditional silica fibre optic lines. This microgravity-produced fibre has numerous applications, including transatlantic telecommunications, high-speed Internet, lasers, as well as enhancing technologies in space."

MIS proved the possibility by launching a miniature fibre-pulling machine called

MIS Fiber, with a payload of ZBLAN preforms supplied by Thorlabs. After being set-up on the ISS, MIS Fiber, remotely operated from Earth, successfully produced high value-to-mass ZBLAN fibre. It returned to Earth with its produce.

As a next step, MIS is now developing Archinaut, which adds robotic arms to their manufacturing set-up so that the parts produced can also be assembled in space. This can be used to build new structures or to repair existing satellites and other space structures.

3D printing in space has some palatable uses too. BeeHex, a start-

Made in Space's Additive Manufacturing Facility (AMF) is the first ever manufacturing service in space (Courtesy: Made in Space Inc.)

